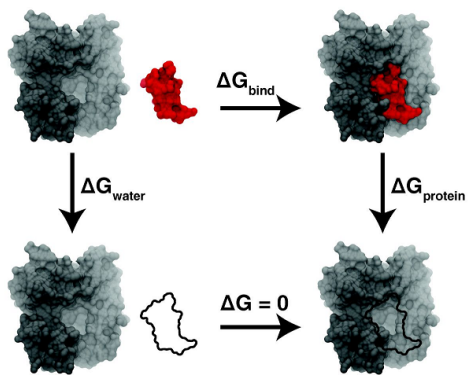


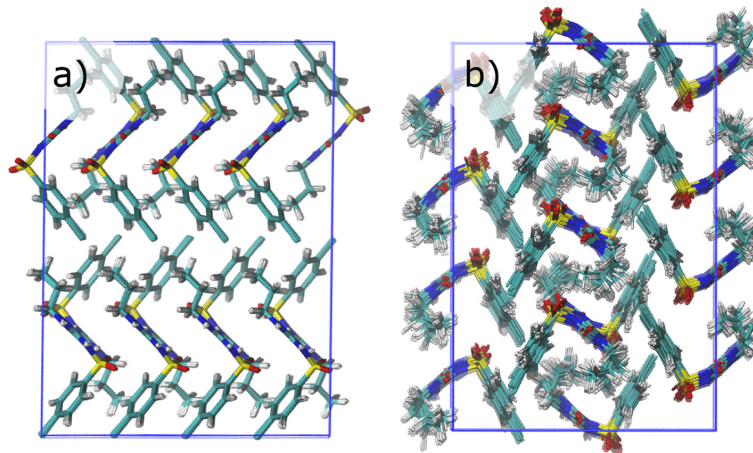
These simple force fields are widely-used tools in computational biophysics and soft matter systems

- Surprisingly, all-atom fixed charge force fields strike an appropriate balance between chemical accuracy and computation efficiency for many systems of interest.

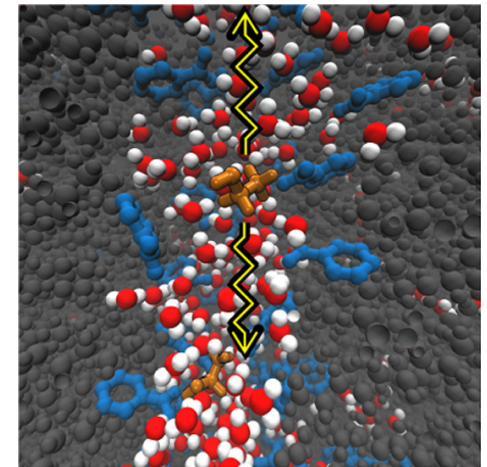
Protein-Ligand Binding



Drug molecule screening



Molecular transport

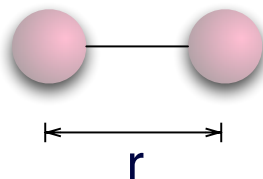


Bonds and Angles

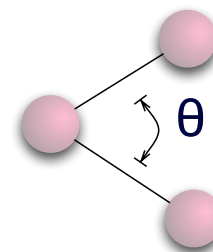
Harmonic bonds and angles

Pretty good approximation to quantum mechanics!

$$+ \sum_{\text{all bonds}} \frac{1}{2} K_b (r - r_0)^2$$

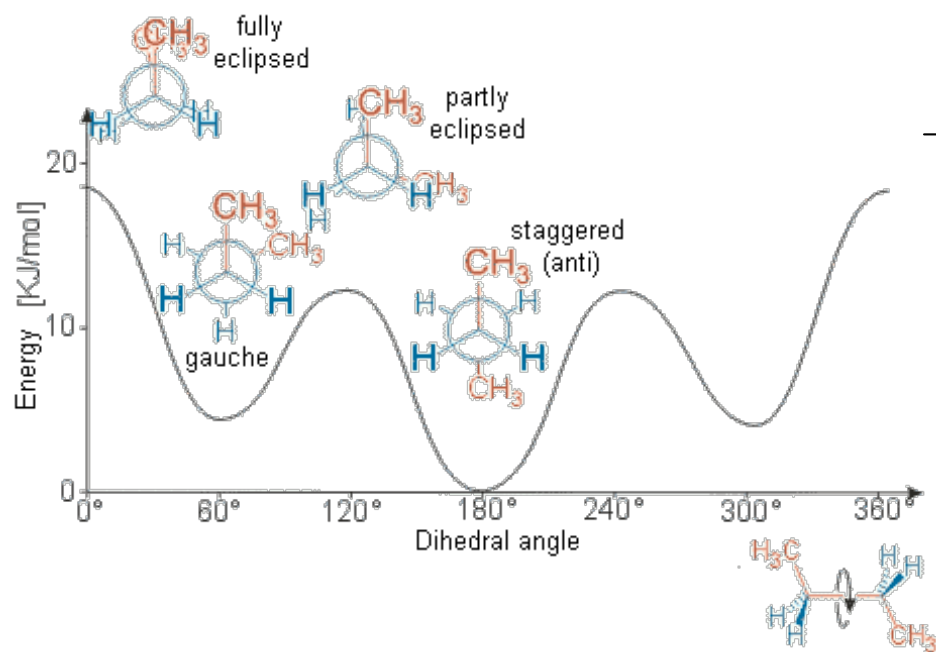


$$+ \sum_{\text{all angles}} \frac{1}{2} K_\theta (\theta - \theta_0)^2$$

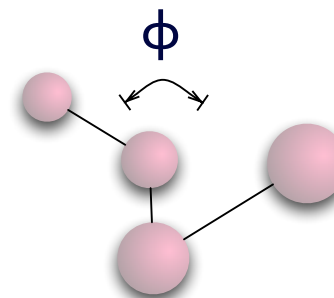


Torsions, or the junk bin of force fields

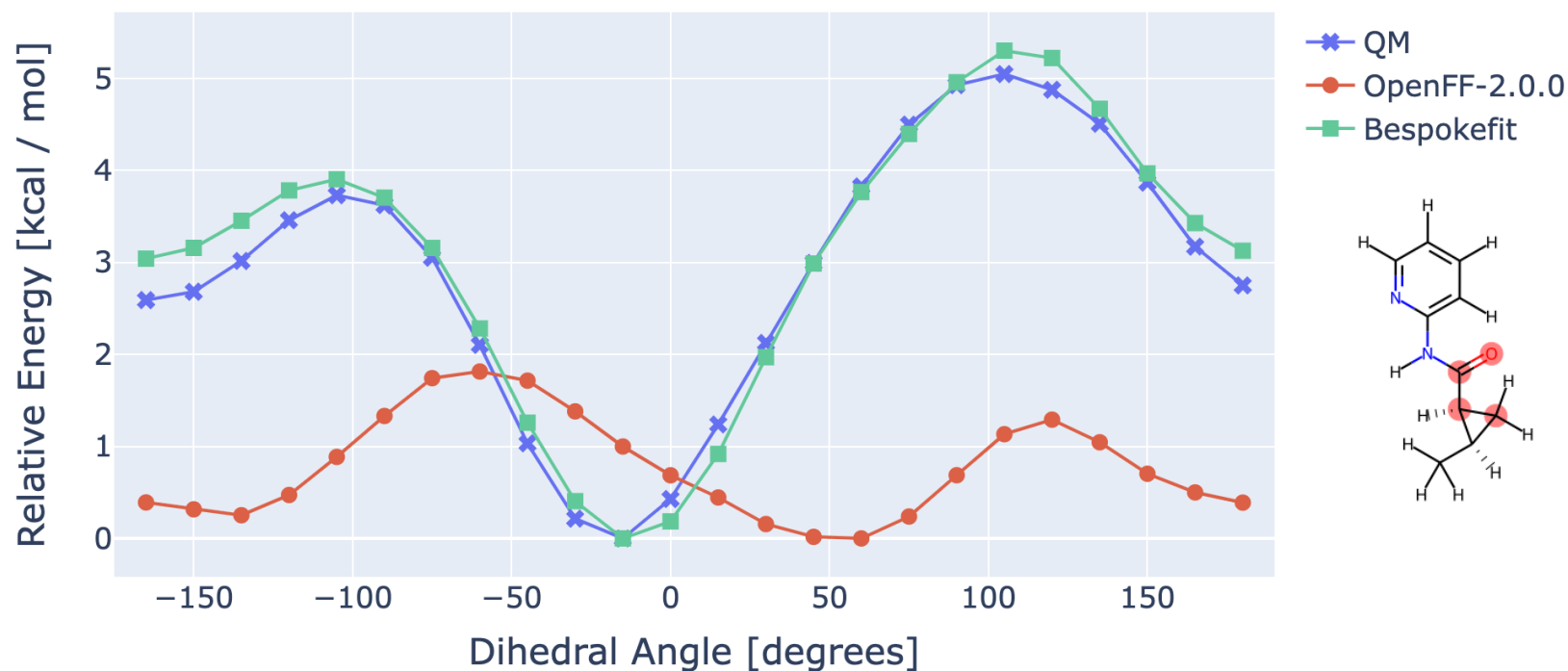
- Replace QM with a sum of periodic trigonometric functions



$$+ \sum_{\text{all torsions}} \sum_n K_\phi [1 - \cos(n\phi + \phi_0)]$$

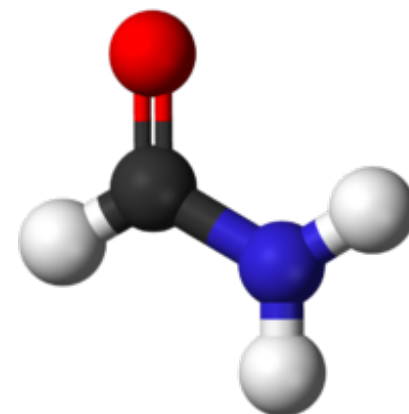
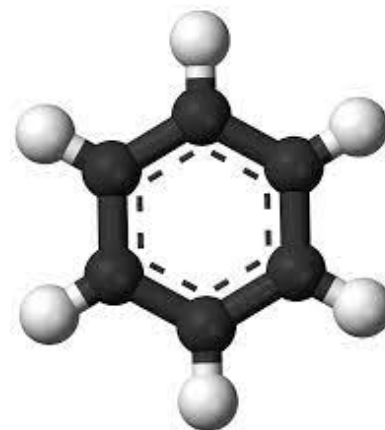
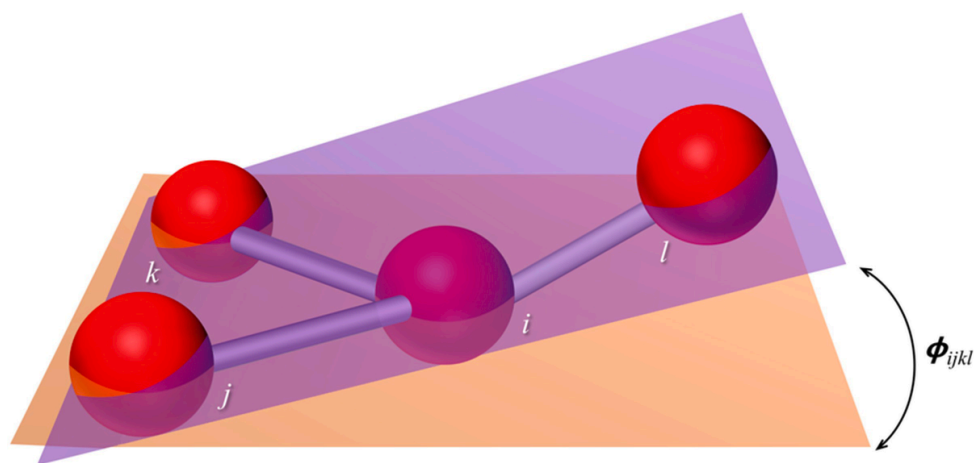


It can be difficult to get these right!



Improper torsions

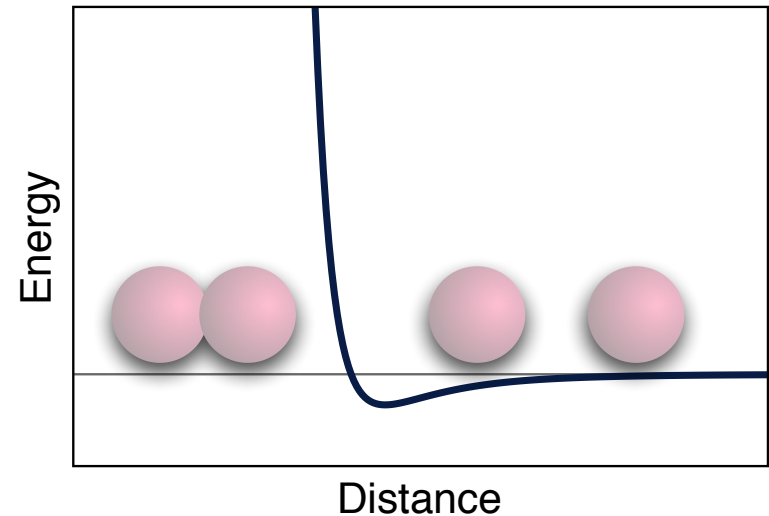
- Basically, a way to keep conjugated systems planar



Let's look at a GROMACS topology file

Lennard-Jones terms

$$U(r_{ij}) = 4\epsilon \left(\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right)$$



```
[ atomtypes ]  
; atom type name type      mass      sigma      epsilon  
opls_111      OW      15.99940    0.000      A      3.12171e-01  7.94960e-01  
opls_112      HW       1.00800    0.000      A      0.00000e-01  0.00000e-01
```

- This just has sigma and epsilon for O and H. What about the O-H interaction???

Combining rules

```
[ defaults ]  
; nbfunc      comb-rule      gen-pairs      fudgeLJ  fudgeQQ  
1             3              yes              0.5      0.5
```

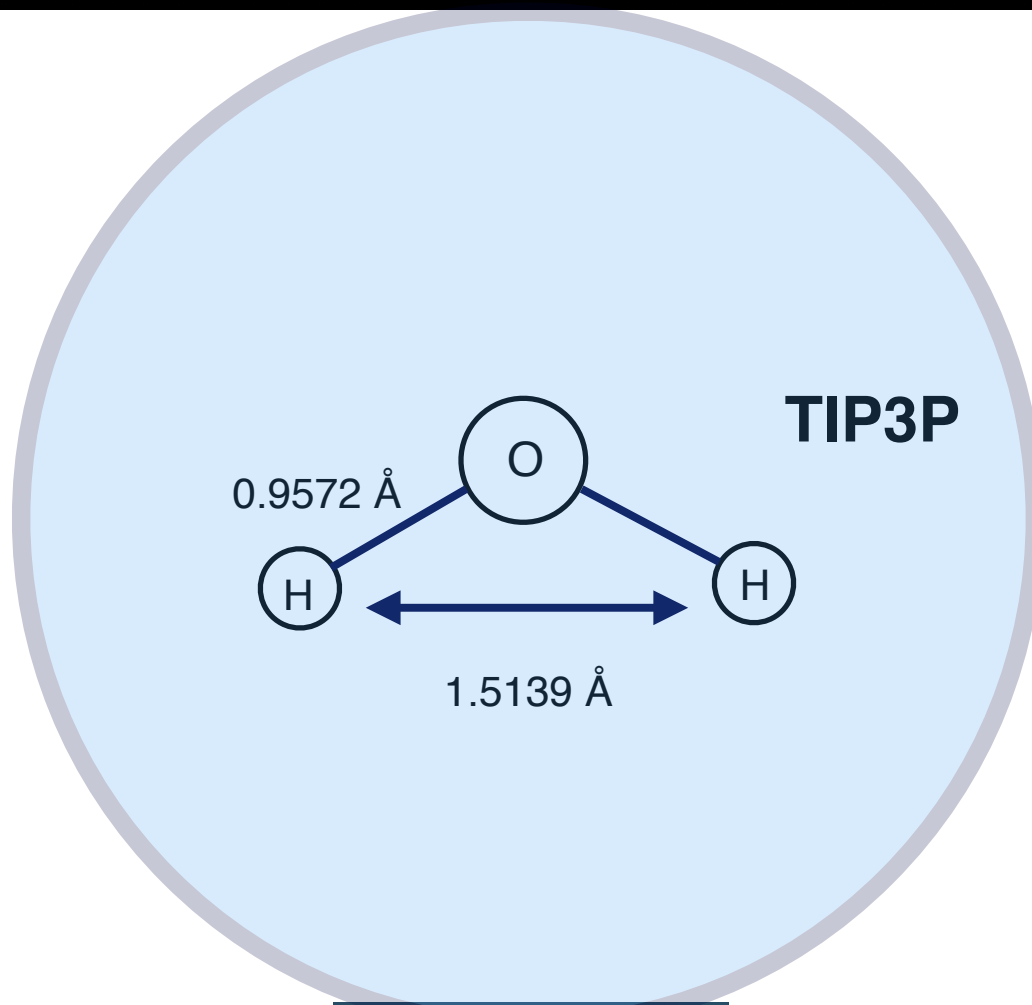
Geometric $\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}$ $\sigma_{ij} = \sqrt{\sigma_i \sigma_j}$

Lorentz-Berthelot $\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}$ $\sigma_{ij} = \frac{1}{2} (\sigma_i + \sigma_j)$

Works better than it should (but not perfect)

Hydrogens and LJ terms

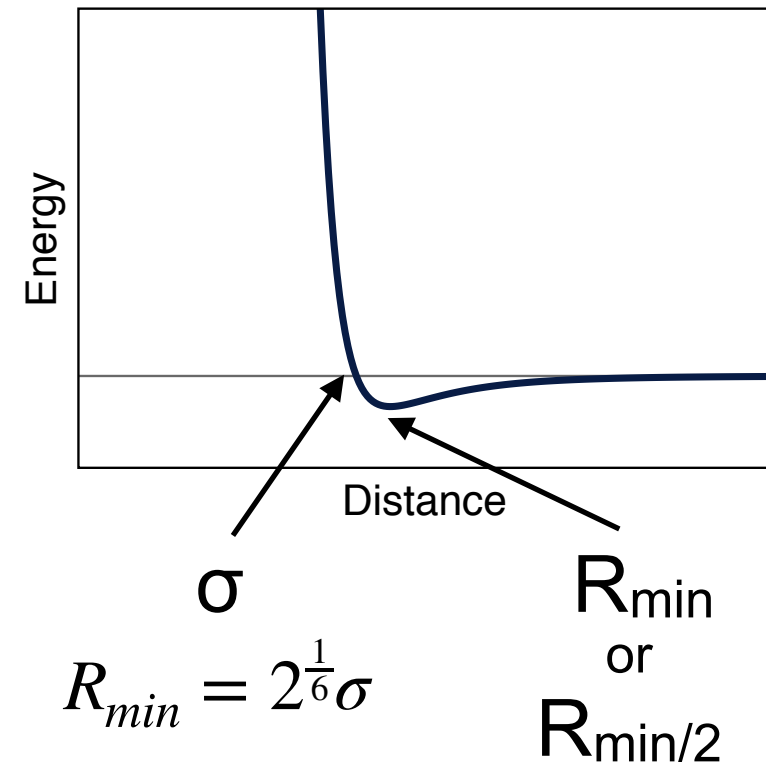
```
[ atomtypes ]  
; atom type name type      mass      0.000      A      sigma      epsilon  
opls_111      OW      15.99940      0.000      A      3.12171e-01      7.94960e-01  
opls_112      HW      1.00800      0.000      A      0.00000e-01      0.00000e-01
```



Lots of different, equivalent ways to write Lennard-Jones parameters!

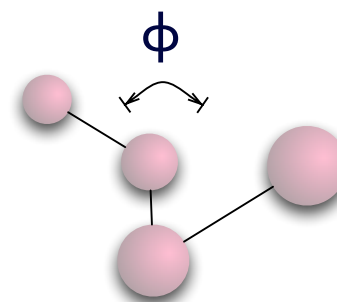
$$U(r_{ij}) = 4\epsilon \left(\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right)$$

$$U(r_{ij}) = \frac{A_{ij}}{r^{12}} - \frac{B_{ij}}{r^6}$$



Exclusions?

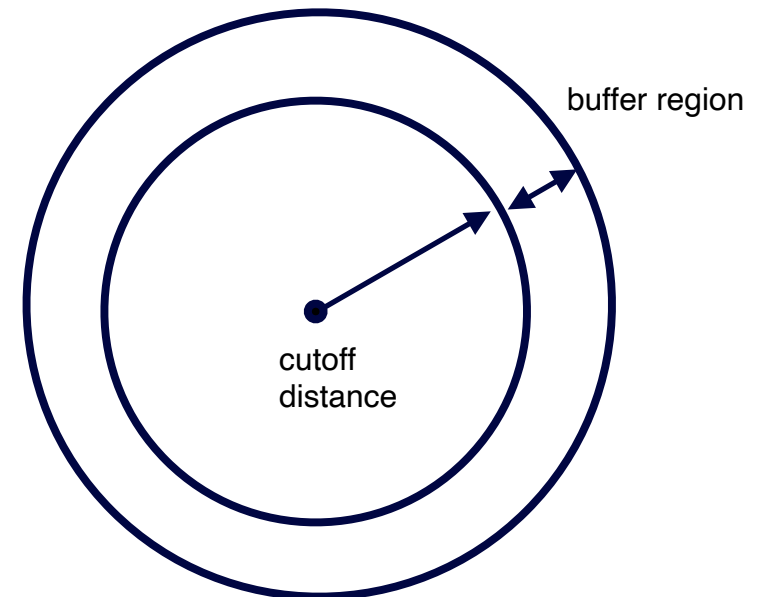
- What about atoms separated by no bonds (1,2)?
- What about atoms separated by 1 bond (1,3)?
- What about atoms separated by 2 bonds (1,4)?



```
[ defaults ]  
; nbfunc      comb-rule      gen-pairs      fudgeLJ  fudgeQQ  
1             3              yes              0.5      0.5
```

Handling Long Range Coulomb Interactions

- Do you calculate the pairwise interaction between ALL atoms in the system?
- Handle short range and long range interactions differently.
- For short range interactions, just keep track of all of the ones that are CLOSE to the cutoff
- For short range interactions, just keep track of all of the ones that are CLOSE to the cutoff
- Update less frequently
- Called Verlet pair lists

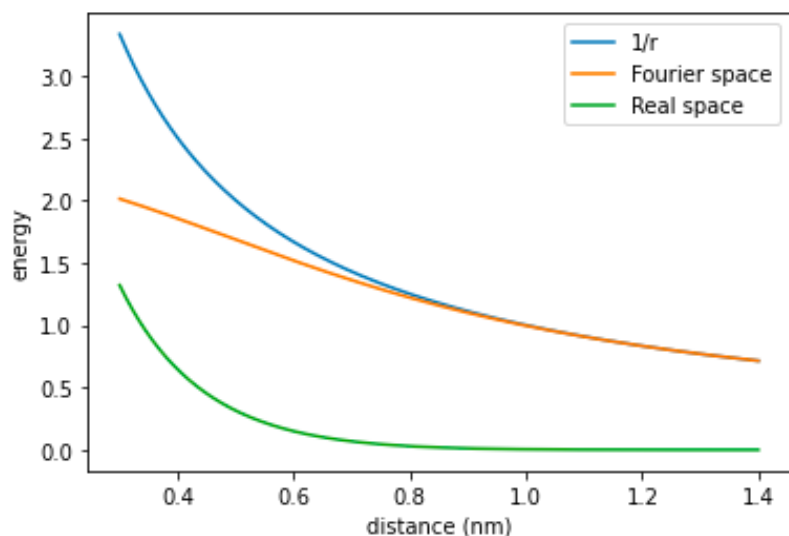


Options for handling the long range interactions

- Direct cutoff
 - Ignore
 - Doesn't work, because Coulomb's law ($U \propto \frac{q_1 q_2}{r}$) decays slowly
- Reaction field
 - Ignore, using some corrections with dielectric theory
 - OK for uncharged systems sometimes
- Ewald summation
 - Treat with Fourier analysis
 - Made possible by the periodic boundary conditions
 - infinite periodic systems can be treated well with periodic functions like sine and cosines

Ewald summation splits energy into two parts

$$U(r) = a \frac{1}{r} = a \left[\frac{1 - \text{erf}(\alpha r)}{r} + \frac{\text{erf}(\alpha r)}{r} \right]$$



- α term controls how rapidly the real space term decreases with increasing r
- Pushes energy back and forth between **real space** and **Fourier space**
- Allows us to tune the cutoff and reduce the amount of work done in real space

- Ewald summation via PME works really well
 - You probably don't need to change the defaults in most cases.

Long range van der Waals interactions

- Options:
 - Direct cutoff
 - Analytical correction
 - Ewald summation (works for $1/r^6$ as well as $1/r$!)
 - Analytical correction works great for radially homogeneous systems
 - assume that the radial distribution function is 1 outside of the cutoff
- $$E = \sum_{\text{pairs}} \int_0^{\infty} -\epsilon \left(\frac{\sigma}{r} \right)^6 4\pi r^2 dr$$
- Ewald summation works as well, but there's not yet consensus about whether to use or not for all simulations, as it adds expense (5-10%)

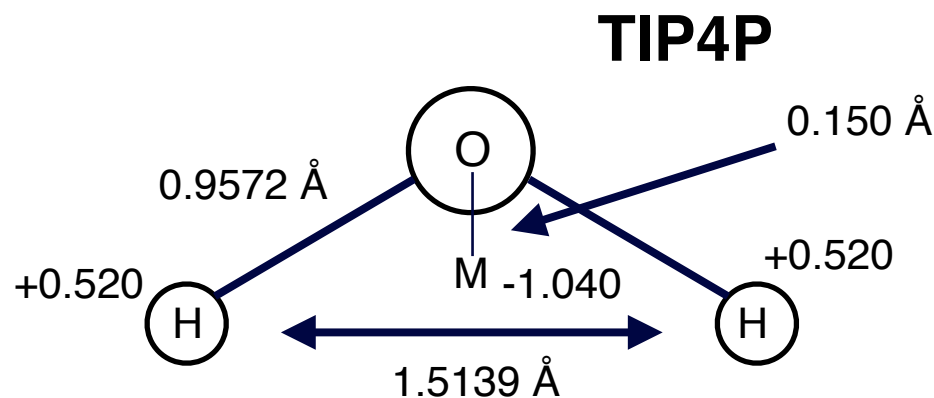
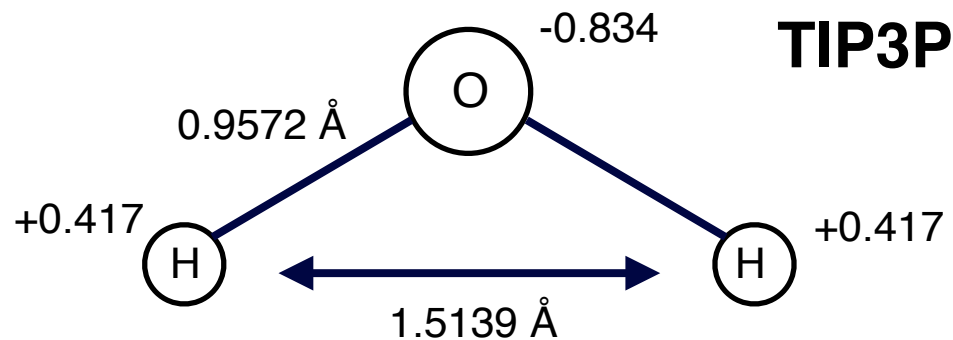
Constraints

- If a harmonic term has as too high a frequency, it requires a shorter time step
- The FASTEST harmonic terms in MD are:

- Anything involving a hydrogen: $\omega = \sqrt{\frac{k}{m}}$
- We FREEZE these bonds rigidly
- If we have a diatomic molecule or rigid triangle (water!), this freezing can be done analytically
- If we have a more complex molecule, this freezing must be corrected for **iteratively**
- It's not really unphysical for water: the vibrational ground state of water is near $P=1$ at room temperature.

Virtual Sites

- Sometimes we want to put charge where there is NO atom present



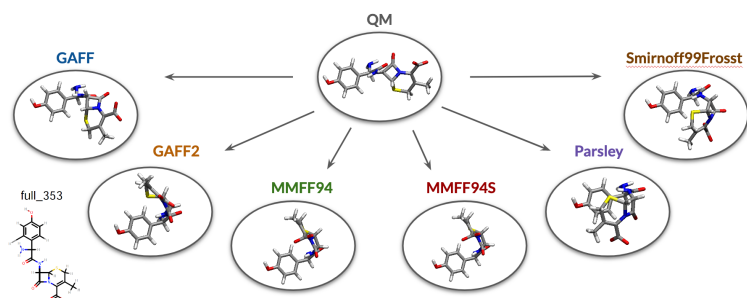
- Point M has no mass or van der Waals term
- Point M is determined at each MD step by the geometry of the other points
- Forces on point M are decomposed into the other particles with actual mass
- Could be interpreted as a force on massive particles with terms more complicated than $1/r$

How do you choose force field parameters??

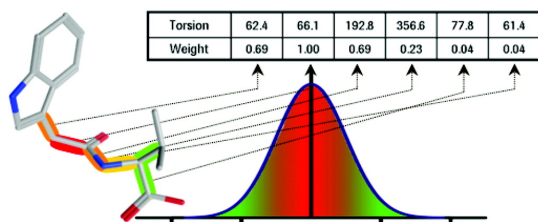
- **Bonded parameters**
- Match them to QM calculations
- Gas phase QM is very quite accurate for matching bonds (lengths and force constants) and angles (equilibrium angles and force constants)
- BUT TORSIONS ARE REALLY HARD!!!!

All the work to do to match force fields for bonded interactions

- 1. Geometry optimization:** Molecular geometries are optimized with various FFs starting from the same QM geometry.



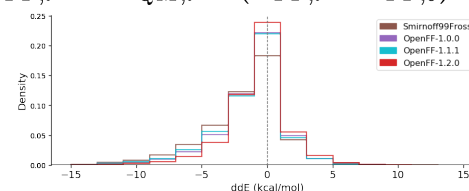
- 3. Geometry comparison:** Geometry is evaluated through root-mean-square deviation (RMSD) and torsion fingerprint deviation (TFD)².



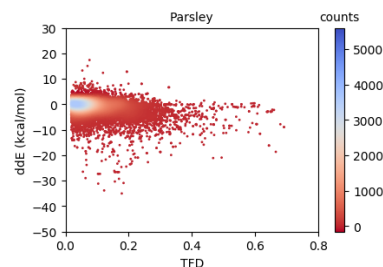
TFD uses Gaussian-weighted differences of torsion angles between two conformations and may be more independent of molecular size for structure comparison purposes.

- 2. Energy comparison:** The ddE for some conformer i is calculated relative to the conformer with the lowest QM energy (0) for different force fields:

$$ddE = dE_{FF,i} - dE_{QM,i} = (E_{FF,i} - E_{FF,0}) - (E_{QM,i} - E_{QM,0})$$



- 4. ddE vs TFD plots:** Compare high density regions of energy vs. geometry data. A perfect match between FF and QM would result with all points at (0, 0).



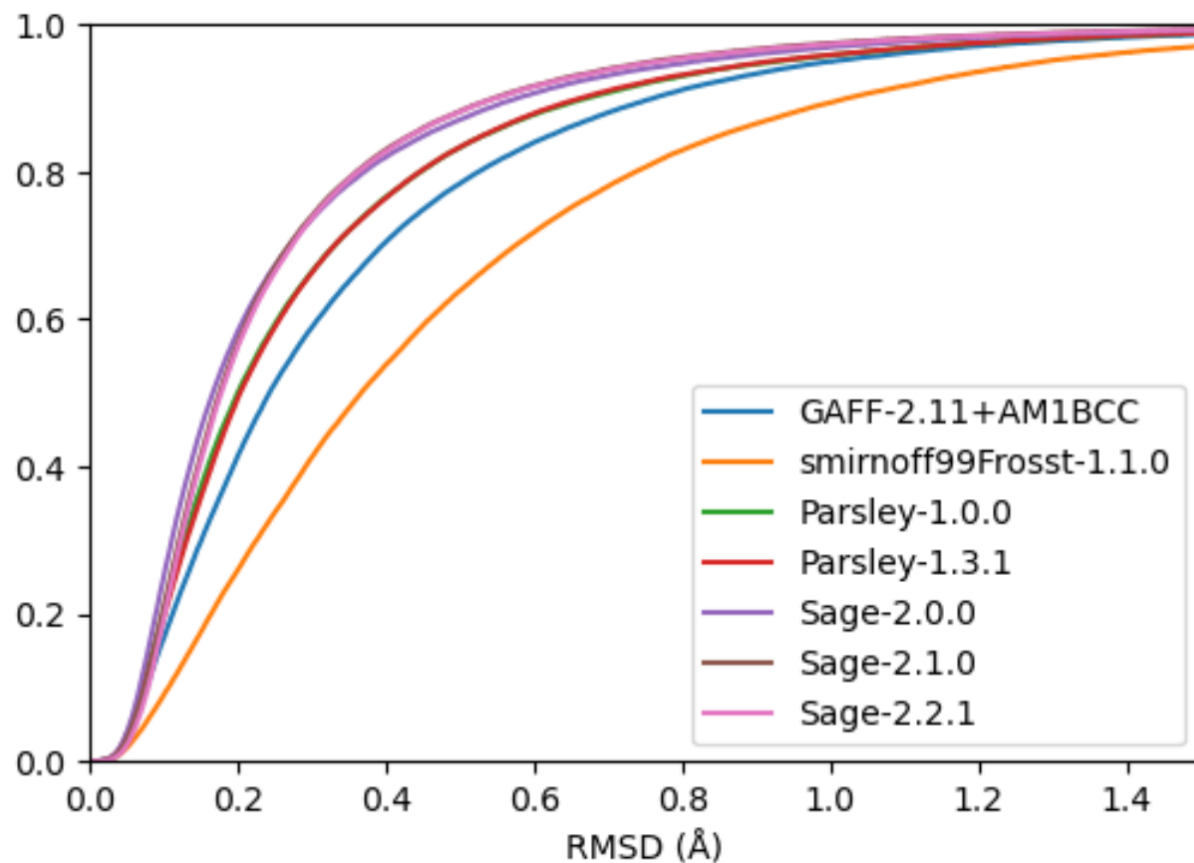
There is not a direct relationship between the accuracy in geometry and the accuracy in relative energies.

1. <https://doi.org/10.26434/chemrxiv.12551867.v2>
 Victoria Lim <https://youtu.be/RZcAL-SSmZU>

2. <https://pubs.acs.org/doi/full/10.1021/ci2002318>

3. https://openforcefield.org/news/benchmark_small_molecules/

Keep tuning force fields to get these quantities better



Cumulative
distribution of
RMSD between QM
and MM minima for
QM datasets

How do you choose force field parameters?

- **Nonbonded parameters**
- **Charges:**
 - Match to electrostatics potential (RESP)
 - With maybe some tweaks
 - AM1 with bond charge corrections (AM1-BCC)
- **van der Waals**
 - Really hard to find QM properties to match
 - Instead, match thermodynamic properties of liquids of small molecules
 - Densities, enthalpies of vaporization or mixing + others

For biological systems, do you have to pick your own parameters?

- Generally NO
- Other people have come up with parameters for biological systems
- ways to apply them to structures generates from the workflows Dr. Sprenger is talking about

What sorts of biomolecular force fields are out there?

- OPLS-AA / OPLS-UA
 - Small molecules, proteins
 - Evolved out of Bill Jorgensen's work
 - Now run by Schrödinger
 - Biomolecules and small molecule ligands

Other Force Fields

- CHARMM
 - Evolved from the work in Martin Karplus's group at Harvard over ~50 years.
 - Alex MacKerrell (Maryland) and Benoit Roux (Chicago) are longest contributing developers but MANY more.
 - Proteins, lipids, nucleic acids small molecules
 - Particularly good for proteins and lipids

Other Force Fields

- AMBER
 - Evolved from the work in Peter Kollman's group at UCSF, and his academic descendants.
 - GAFF small molecule force field (Junmei Wang)
 - Proteins (ff99sb, ff14sb, ff19sb), nucleic acids, and carbohydrates (Roitberg and Simmerling)
 - Ions (Jeong and Cheatham)

Other Force Fields

- Open Force Field
 - <http://www.openforcefield.org>
 - New effort, focused on open software as well as open force fields
 - Small molecule based, in the process of adding biopolymers
 - Can build from SMILES, sdf, mol2
 - Difference is: it matches molecular parameters in a more general way and DOESN'T use atom types



Other Force Fields

- What about polymers?
 - TRAPPE
 - Coming mostly from Ilja Siepmann's group
 - Designed for phase equilibrium
 - COMPASS
 - MMFF
- What about metals?
 - Lots of metal specific functional forms
 - Generally requires more complicated interactions than the simple nonbonded potentials
 - For bulk metals, these are generally ATOMIC force fields.

Force Field Assignment Tools

- People who developed force fields made tools to apply them to chemical systems
 - Proteins
 - Small molecule ligands
 - Nucleic acids
- Amber force fields = AmberTools/tleap
- CHARMM force fields = CHARMM-GUI/CGENFF
- OpenFF forcefields = OpenFF toolkit
- foyer (MoSDeF ecosystem) = polymer systems

But What Force Field Should I Use?

- The one that allows you to test the hypothesis that you want to

OR

- The one that is validated as an effective screening mechanism to predict the property of interest to other researchers.
- Has the force field I want to use been validated on sufficiently similar systems?
- "All models are wrong. Some models are useful."